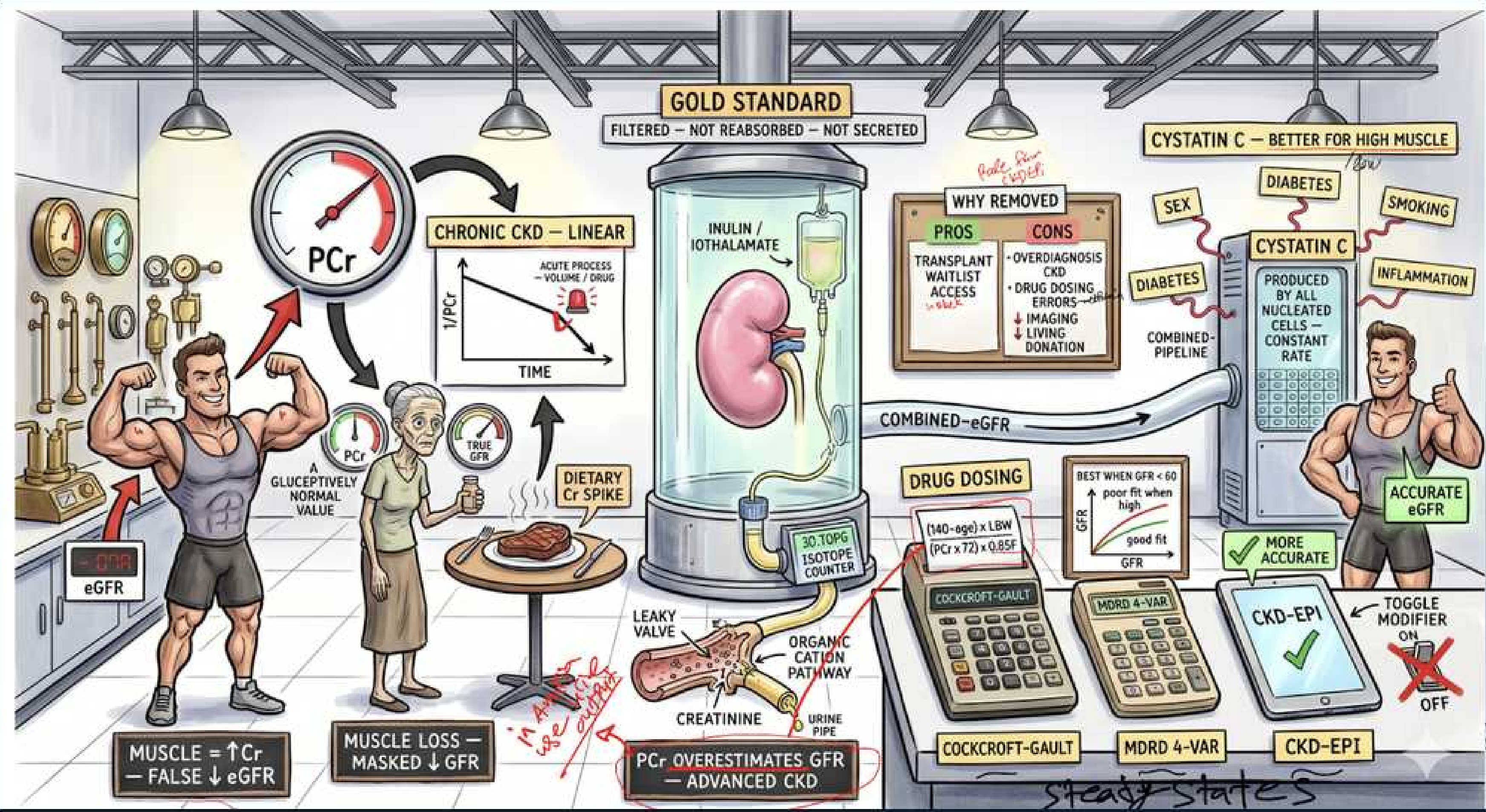
Estimating GFR — Creatinine, Cystatin C & eGFR Equations



1. Gold standard — inulin / iothalamate

WHAT YOU SEE

Gold standard tank, inulin/iothalamate

WHAT IT ENCODES

True GFR is measured by clearance of an ideal marker (inulin or iothalamate): freely filtered, not reabsorbed, not secreted. Too cumbersome for clinical use, so we ESTIMATE GFR from endogenous markers (creatinine, cystatin C).

BOARD TRAP

Wrong answer: creatinine clearance equals true GFR. Discriminator: inulin/iothalamate is the gold standard; creatinine is secreted and overestimates GFR.

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2. PCr vs GFR — non-linear curve

WHAT YOU SEE

PCr vs GFR hyperbolic curve, chronic CKD linear

WHAT IT ENCODES

The serum creatinine-GFR relationship is HYPERBOLIC: early in disease large GFR drops cause small creatinine rises, so creatinine is insensitive to early renal impairment. Only at low GFR does creatinine rise steeply.

BOARD TRAP

Wrong answer: a normal creatinine means normal GFR. Discriminator: due to the hyperbolic curve, GFR can fall ~50% before creatinine leaves the normal range.

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3. Acute vs chronic creatinine kinetics

WHAT YOU SEE

Acute process: volume/drug spike

WHAT IT ENCODES

In ACUTE kidney injury, creatinine has not yet equilibrated, so a given creatinine underestimates the severity of GFR loss; eGFR equations assume steady state and are INVALID in AKI. Use trends, not a single eGFR.

BOARD TRAP

Wrong answer: apply CKD-EPI eGFR during acute rising creatinine. Discriminator: eGFR equations assume steady state — they are invalid in non-steady-state AKI.

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4. Muscle mass — high Cr, false low eGFR

WHAT YOU SEE

Muscular figure, muscle = high Cr

WHAT IT ENCODES

Creatinine derives from muscle, so high muscle mass (bodybuilders) raises creatinine independent of GFR → falsely LOW eGFR. Creatine supplements and high meat intake similarly raise it.

BOARD TRAP

Wrong answer: high creatinine in a muscular patient means CKD. Discriminator: muscle mass elevates creatinine non-renally — consider cystatin C for accuracy.

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5. Muscle loss — masked low eGFR

WHAT YOU SEE

Elderly/cachectic figure, muscle loss masks low GFR

WHAT IT ENCODES

Low muscle mass (elderly, cachexia, cirrhosis, amputees) lowers creatinine production, so a normal creatinine can MASK significant GFR loss → falsely HIGH eGFR and overdosing of renally-cleared drugs.

BOARD TRAP

Wrong answer: normal creatinine in a frail elder excludes CKD. Discriminator: low muscle mass falsely normalizes creatinine; cystatin C is more reliable.

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6. Dietary creatinine spike

WHAT YOU SEE

Cooked meat at table, dietary Cr spike

WHAT IT ENCODES

A cooked-meat meal transiently raises serum creatinine (preformed creatinine), independent of renal function. Fast or use steady-state values when interpreting borderline creatinine.

BOARD TRAP

Wrong answer: a post-meal creatinine rise means worsening renal function. Discriminator: dietary meat causes a transient non-renal creatinine bump.

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7. Tubular secretion — drugs block it

WHAT YOU SEE

Organic cation pathway, leaky valve

WHAT IT ENCODES

Creatinine is freely filtered AND secreted by proximal organic cation transporters, so creatinine clearance OVERESTIMATES true GFR. Cimetidine, trimethoprim, and others block secretion → creatinine rises with no true GFR change.

BOARD TRAP

Wrong answer: a creatinine rise on trimethoprim/cimetidine means AKI. Discriminator: these block tubular secretion of creatinine — true GFR is unchanged.

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8. PCr overestimates GFR in advanced CKD

WHAT YOU SEE

PCr overestimates GFR, advanced CKD

WHAT IT ENCODES

As GFR falls, tubular secretion contributes a larger fraction of creatinine clearance, so creatinine clearance increasingly OVERESTIMATES true GFR in advanced CKD. Equations and cystatin C help correct this.

BOARD TRAP

Wrong answer: creatinine clearance underestimates GFR in CKD. Discriminator: secretion makes it OVERESTIMATE GFR, more so as CKD advances.

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9. Cystatin C — better for high muscle

WHAT YOU SEE

Cystatin C, better for high muscle

WHAT IT ENCODES

Cystatin C is produced by all nucleated cells at a constant rate and is independent of muscle mass, age, and sex — better at extremes of body composition. Combined creatinine-cystatin C eGFR is the most accurate estimate.

BOARD TRAP

Wrong answer: cystatin C is unaffected by anything. Discriminator: cystatin C is raised by steroids, thyroid disease, smoking, and inflammation — not a perfect marker.

10. Cystatin C confounders

WHAT YOU SEE

Steroids, thyroid, smoking, inflammation, diabetes

WHAT IT ENCODES

Cystatin C is increased by corticosteroids, hyperthyroidism, smoking, systemic inflammation, and diabetes independent of GFR. Know these confounders before relying on cystatin-based eGFR.

BOARD TRAP

Wrong answer: cystatin C cannot be confounded. Discriminator: steroids/thyroid/inflammation raise cystatin C, mimicking lower GFR.

11. Cockcroft-Gault — drug dosing

WHAT YOU SEE

Cockcroft-Gault calculator, drug dosing

WHAT IT ENCODES

Cockcroft-Gault estimates creatinine clearance (uses age, weight, sex) and historically anchors many drug-dosing labels. It is not body-surface-area normalized and tends to be less accurate than CKD-EPI for staging.

BOARD TRAP

Wrong answer: use CKD-EPI for all renal drug dosing. Discriminator: many drug labels are validated against Cockcroft-Gault creatinine clearance.

12. MDRD vs CKD-EPI

WHAT YOU SEE

MDRD 4-var and CKD-EPI calculators

WHAT IT ENCODES

MDRD underestimates GFR at higher values; CKD-EPI is more accurate across the range and is the preferred staging equation. The 2021 CKD-EPI equation removed the race coefficient.

BOARD TRAP

Wrong answer: MDRD is preferred for staging. Discriminator: CKD-EPI is more accurate, especially at near-normal GFR, and is the current standard.

13. CKD-EPI — most accurate eGFR

WHAT YOU SEE

CKD-EPI checkmark, accurate eGFR

WHAT IT ENCODES

CKD-EPI (creatinine, or combined creatinine-cystatin C) is the recommended eGFR equation for CKD classification. The combined cystatin-creatinine version is the most accurate, especially when muscle mass is atypical.

BOARD TRAP

Wrong answer: any single equation suffices in all patients. Discriminator: confirm with cystatin C / combined CKD-EPI when muscle mass or the clinical picture is discordant.

14. 2021 race-free equation

WHAT YOU SEE

Toggle modifier OFF, race removed

WHAT IT ENCODES

The 2021 CKD-EPI refit removed the race coefficient to avoid inequitable GFR estimates. Cystatin C-based and combined equations are now favored to provide race-independent, accurate estimates.

BOARD TRAP

Wrong answer: still apply a race correction factor. Discriminator: the 2021 CKD-EPI equations are race-free; combined cystatin C improves accuracy.